A guide for the warehouse team

Warehouse Planning Agent

An AI agent that plans a footwear warehouse's day: how much will ship, which SKUs are in demand, which boxes to bring down from levels 3-4, and how many workers each day needs.



Levels 3-4 hold reserve. Levels 1-2 are within reach.

The agent says which boxes come down.

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shreyasthakur11.github.io/warehouse-planning-agent

Start here

Four answers before picking starts

Every morning a warehouse needs to know how much will ship, which SKUs will sell, which boxes to bring within reach and how many people to bring in. This agent works out all four from the order history, checks its own plan and writes a brief for the supervisor. This guide shows how to read that plan and how far to trust it.

1. Demand

How much will ship?

Pairs, order lines and orders for each of the next 5 working days, with a likely range.

2. SKUs

Which SKUs are in demand?

The 15 SKUs expected to sell most, with their trend and where their stock sits.

3. Replenishment

What comes down from levels 3-4?

Every box to move down to levels 1-2, so pickers can reach enough stock for the next 2 working days.

4. Workers

How many workers each day?

Hours of picking and moving, turned into workers, with the number to bring in if the day runs high.

What it is

- An AI agent that runs the morning's planning on its own and shows every step it took.
- Tested on 134 working days of real records, each planned from earlier orders only.
- A live dashboard that anyone can open in a browser.

What it is not

- It does not run the warehouse. A person carries out the plan.
- It is not connected to a live system. It replays records that end on 19 October 2023.
- It never makes up a number. Every figure comes from a tool.

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The idea

What makes it an agent

An ordinary program runs the same steps every time. An agent is given a goal and a set of tools. It decides which tool to use next, reads the answer and decides what to do with it, until the goal is met. It also knows when to stop and ask a person.

It works in a loop

Observe
Read the orders and grade the last plan.

Decide
Choose the forecast method that missed least, and how to staff.

Act
Forecast, find the SKUs in demand, plan the boxes and the workers.

Check
Test the plan against the roster and the racks.

Report
Write the brief and raise what a person must decide.

Its 8 tools

Tool	What it returns
`read_orders`	How much history there is, and the last working day's volume
`check_yesterday`	The last plan's forecast against what shipped
`compare_forecast_methods`	Each method's miss over the last 30 working days
`forecast_demand`	Pairs, lines and orders for the next 5 working days, with a range
`find_skus_in_demand`	The 15 SKUs expected to sell most, with trend and stock
`plan_replenishment`	Every box to bring down from levels 3-4 to 1-2
`plan_workers`	Hours of work and workers for each forecast day
`check_plan`	What a person should know or decide

Two brains, one set of tools

The built-in rules

Run the loop with no key and no cost. Every score in this guide comes from them.

Claude

With an Anthropic API key, Claude runs the same loop through the same tools. The dashboard says which brain made each plan.

> **The language model never calculates a number**
>
> Tools do every sum. A brief that states a number no tool produced is rejected, and the brief is written from the tools' answers instead.

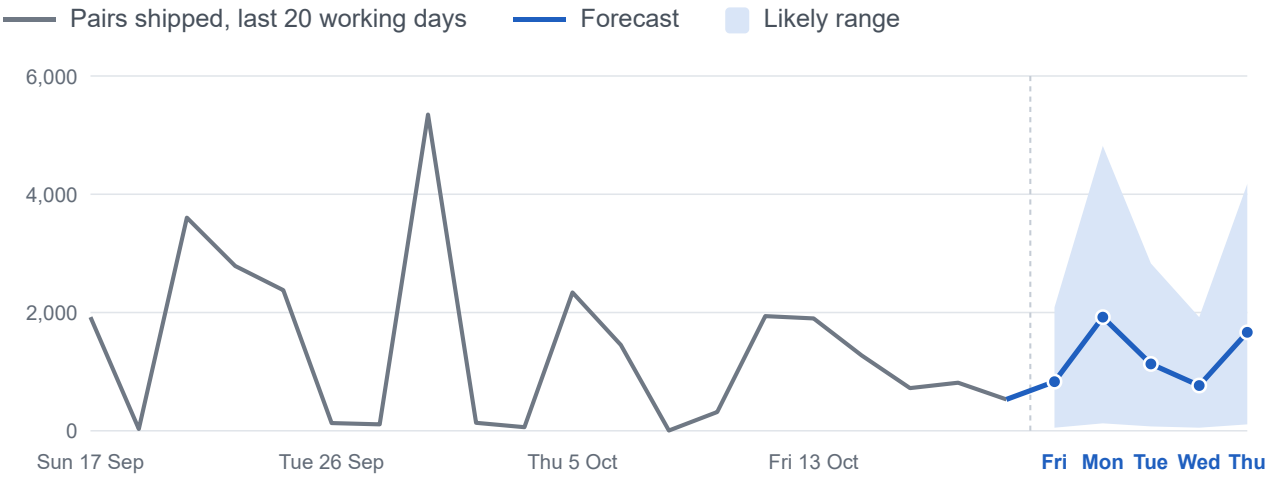
How much will ship

The worked example is the plan for Friday 20 October 2023, the first working day after the records end. Like every plan on the dashboard, it opens with the agent's brief.

> **THE AGENT'S BRIEF**
>
> Friday 2023-10-20: expect about 832 pairs (54 to 2,085) in 471 order lines. Bring down 62 boxes from levels 3-4 to 1-2 before picking starts, starting with AX7BJ8 size 9 from M-24-30. Staff 1 worker, 3 if the day runs high. Rising: AKP0A7.

832	471	1	62
pairs expected, likely 54 to 2,085	order lines, in about 110 orders	worker, 3 if the day runs high	boxes to bring down from levels 3-4 to 1-2

The forecast



How it chose the method

Before forecasting, it scores three methods on the last 30 working days and uses the one that missed least.

Method	Missed by, last 30 days
Average of the last six same weekdays	**70.8%**
Last 20 days' average, shaped by weekday	71.7%
Same weekday last week	84.2%

The range spans the middle 80% of this method's misses over those days. A wide range means volume has been swinging, not that the arithmetic is unsure.

SKUs, and the boxes to bring down

It shares the forecast out across SKUs by their sales over the last 40 working days, then checks that each has enough boxes within a picker's reach.

SKUs in demand

SKU	Pairs, next 5 days	Share	Lately	Boxes, levels 1-2	Levels 3-4	Days levels 1-2 last
8N10W9	551	8.7%	Steady	541	51	7.5
05W6TK	484	7.7%	Falling	443	59	6.9
I1KDJ0	288	4.6%	Steady	429	56	11.3
WRRW1W	251	4.0%	Steady	572	61	17.3
SMMRK3	223	3.5%	Steady	597	50	20.3
PY5UPB	222	3.5%	Falling	156	52	5.3

The top 15 take 50.5% of the forecast. Lately compares the last 10 days with the SKU's usual rate.

Bring down from levels 3-4 to 1-2

For each SKU size, it compares what levels 1-2 hold with what the next 2 working days need, and lists the boxes to bring down, the ones about to run out first. Below are the first three on the list, then the three largest top-ups.

62
boxes to bring down

32
SKU sizes

7
new pick faces

23 Oct
covered until, Monday

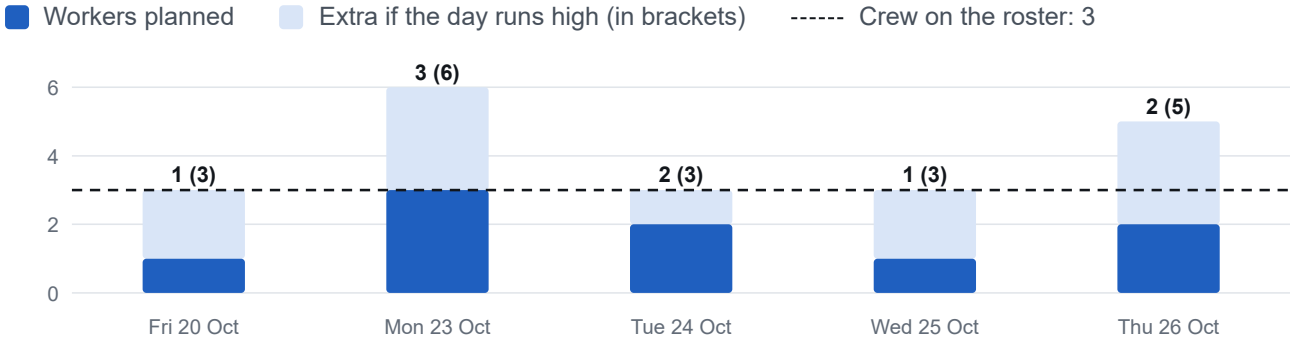
SKU size	Boxes	From	To	Why
AX7BJ8 size 9	1	M-24-30 L3	M-22-13 L1	New pick face
EUMBM7 size 9	1	Q-21-30 L3	Q-21-12 L1	New pick face
5MNTT7 size 11	1	E-14-30 L3	E-14-11 L1	New pick face
9FWH2G size 9	7	A-22-30 L3 and 6 more	F-21-13 L1	4 on L1-2, 13 needed
DSQXYR size 8	5	F-13-30 L3 and 4 more	A-20-23 L2	4 on L1-2, 9 needed
9FWH2G size 11	4	K-02-30 L3 and 3 more	B-24-21 L2	1 on L1-2, 11 needed

A new pick face needs a free slot, so it sends 7 idle boxes up to levels 3-4 to make room. It also notes what it cannot fix: 6 SKU sizes will run short by about 24 pairs, and 138 SKU sizes expected to sell have no box in the racks.

Workers, and the supervisor's list

It turns each day's forecast into hours of work, then workers. It staffs to the top of the range when 4 of the last 5 days ran above forecast. Here 2 did, so it staffed to the expected volume.

Workers each day



Day	Picking hours	Moving boxes	Total hours	Workers	If the day runs high
Fri 20 Oct	4.9	1.0	5.9	1	3
Mon 23 Oct	11.2	2.4	13.6	3	6
Tue 24 Oct	6.6	1.4	8.0	2	3
Wed 25 Oct	4.5	1.0	5.4	1	3
Thu 26 Oct	9.7	2.1	11.8	2	5

The time standard: 10 seconds a stop, 6 seconds a pair, 60 seconds to set up each trolley run and 60 seconds to bring a box down, plus a 15% allowance. A worker gives 6.8 productive hours of an 8-hour shift.

The supervisor's list

The last tool checks the plan. Anything it finds goes to the supervisor. Two kinds need a decision before the plan is settled.

Mark	What it found	What to do
ASK	More workers needed than the 3 on the roster	Arrange cover
ASK	Demand the racks cannot cover	Call stock in from production
CHECK	A SKU selling at 3 times its usual rate	Look for one large order
CHECK	Weekly: SKU sizes to sell with no box in the racks	Give them a slot
CHECK	Weekly: boxes on levels 1-2 unsold for 40 days	Move them up to levels 3-4

The plan for Fri 20 Oct raised nothing. The next page shows a day that did.

Checking the plan

How a day went, and the track record

For every day in the records, the dashboard puts the plan beside what happened. Tuesday 11 April 2023 is one of the closer days. The chart below shows the rest.

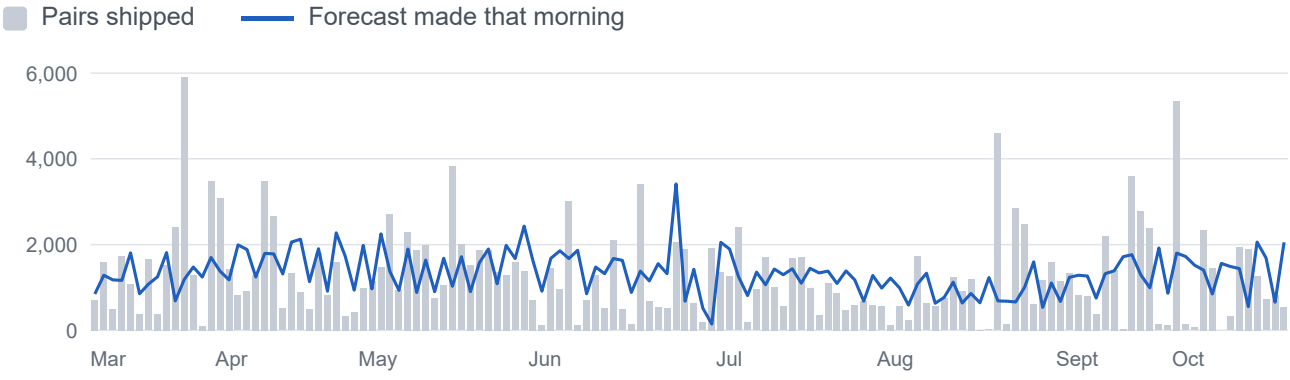
Tue 11 Apr	Planned	Happened
Pairs	1,243	1,284
Order lines	580	964
Workers	2	2 needed
Boxes brought down	144	61 needed

Pairs landed +3% from the forecast, inside the range of 371 to 2,652. Order lines ran above the usual ratio, and the workers planned still covered the day. 61 of the 144 boxes brought down were needed within the 2 days.

What it raised that morning

CHECK	B24J86 has sold at least 3 times its usual rate over the last 10 days. Check whether one large order is behind it.
CHECK	JHSQHS has sold at least 3 times its usual rate over the last 10 days. Check whether one large order is behind it.
CHECK	Weekly review: 131 SKU sizes expected to sell before 2023-04-12 have no box in the racks, about 276 pairs. Give them a slot or pick them where they are kept.
CHECK	Weekly review: 6,112 boxes on levels 1-2 have not sold in 40 working days. Moving them up to levels 3-4 would free pick faces for the SKUs that sell.

All 134 graded days, each planned from earlier orders only



64.9%
forecast miss, against 75.9% for the same weekday last week

79.9%
of days with enough workers, 94.8% when staffed for a high day

27.9%
of boxes brought down were needed within the 2 days

Honesty

How far to trust each part

What the track record says about each answer, and what to do about it.

The forecast: better than the simplest rule, still wide

It misses by 64.9% of the pairs shipped, against 75.9% for the same weekday last week, and 76.1% of days land inside the range. Over the whole period it runs +2.3% from the total, so it does not lean one way. Daily volume swings hard here: on 20 of 178 working days, the warehouse shipped at least ten times the day before, or a tenth of it.

So: plan around the range, not only the middle figure.

Workers: slightly over rather than short

Enough workers were planned on 79.9% of days. 27 days were short and 54 were over, at 2.04 workers planned against 1.84 needed on average. Staffing for a high day would have covered 94.8%. The agent chose to on 16 days.

So: on a day when a late order costs more than an idle hour, staff to the high figure.

Replenishment: the weak part, and why

Of 12,610 boxes brought down, 3,514 were needed in time, and they caught 21.6% of the 16,261 that truly had to come down. The cause is measured: even with the next 5 days' total pairs known exactly, sharing them out by recent sales misses badly.

Recent days used	Error by SKU size	Error by SKU
10	97.2%	76.1%
20	93.5%	73.2%
40 (what the agent uses)	**92.9%**	**73.3%**
60	92.6%	73.2%
90	92.6%	73.0%

So: treat the move list as a starting point. A box is cheap to move; a pair missing from levels 1-2 costs a trip up the rack.

What this does not show

- One stock snapshot: every morning starts from the same stock position.
- The 60 seconds a box and the 15% allowance are assumptions.
- The scores are the built-in rules'. Claude is tested with a stand-in model.

Try it

Run it yourself

The dashboard needs nothing installed. Each calendar day it moves on by one working day and shows that day's plan, made only from the orders before it.



Open the dashboard

shreyasthakur11.github.io/warehouse-planning-agent

Today shows the plan and every step. Track record grades every day. How it works lists every number the agent uses.

Run it on a laptop

```
pip install -e .
warehouse-planner plan                     # the next working day, step by step
warehouse-planner plan --date 2023-04-11   # any day in the records
warehouse-planner replay                   # plan every day and score the plans
warehouse-planner site                     # build the dashboard into _site/
```

To let Claude run the morning, install the `claude` extra and set `ANTHROPIC_API_KEY`. With no key, the built-in rules run it.

Change the numbers

Every number the agent uses sits in `src/warehouse_planner/settings.py`. Change one, then run the replay again.

Setting	Value	Setting	Value
Crew on the roster	3	Days of cover on levels 1-2	2
Shift	8 hours	Allowance	15%
Bringing a box down	60 s	Days ahead planned	5
Days used to score methods	30	Fewest days before it plans	40

The records

Real orders from a footwear manufacturer's warehouse in Sherbrooke, Canada: 178 working days from 5 January 2023 to 19 October 2023, 122,370 order lines, 228,195 pairs and 208 SKUs, stored in 41,256 boxes across 2,292 locations: 30,456 on levels 1-2 and 10,800 on levels 3-4.

de Assis, de Paula Ferreira and Ouhimmou (2025), Order Picking Dataset from a Warehouse of a Footwear Manufacturing Company, Mendeley Data, doi.org/10.17632/pf2w725pw3.1, CC BY 4.0. The dataset's creators do not endorse this project. Code: MIT, at github.com/ShreyasThakur11/warehouse-planning-agent.